

SMART IDEA LAB

IoT-Based Smart Attendance & Entry-Exit Monitoring System

What is the product?

A smart lab monitoring system that uses **RFID, dual IR sensors, ESP32 and an RTC** to automatically identify students, detect whether they are entering or leaving the lab, record the time, and maintain attendance records digitally.

How It Works

1. Detect Direction

IR1 → IR2 = Entry

IR2 → IR1 = Exit

2. Scan RFID Card

After movement is detected, the student scans their registered RFID card at the RFID reader.

3. Verify Student

The ESP32 checks the RFID against the registered student details.

4. Record Attendance

The system records the student's **name, RFID, date, entry/exit time and attendance status**.

5. Update Dashboard & Google Sheets

The information is displayed on the local web dashboard and stored in **Google Sheets** for digital record keeping.

6. Detect Anomalies

The system identifies unusual events such as unregistered RFID, RFID scan without movement, movement without RFID verification, duplicate entry, exit without a recorded entry, and unusually short/long visits.

How to Use the Product

ENTRY	Walk through IR1 → IR2 → Wait for Entry Detected → Scan RFID card → Entry recorded.
EXIT	Walk through IR2 → IR1 → Wait for Exit Detected → Scan RFID card → Exit recorded.

User Output

OLED	Immediate entry/exit feedback
Buzzer	Confirmation or alert
Web Dashboard	Live attendance and lab status
Google Sheets	Digital attendance history

Overall Workflow: Person Movement → IR Direction Detection → RFID Identification → Verification → Time Recording → Dashboard & Google Sheets → Anomaly Analysis

Developed using: ESP32 • RC522 RFID • Dual IR Sensors • DS3231 RTC • OLED • Wi-Fi • Google Sheets